Jadavpur University

Department of Electronics and Tele-Communication Engineering

3D Brain Tumor Segmentation using Deep Learning and Supervoxel-Based Graph Cut Optimization

Submitted By:

Niladri Sekhar Mondal

Roll No: 002210701121

Rivuprovo De

Roll No: 002210701125

Supervised By:

Prof. Ananda Shankar Chowdhury

Abstract

Brain tumor segmentation from 3D Magnetic Resonance Imaging (MRI) is a pivotal task in medical diagnostics and treatment planning. The complexity of tumor structures, combined with the low contrast between healthy and pathological tissues, makes manual segmentation tedious and prone to inter-observer variability. While Convolutional Neural Networks (CNNs), particularly the 3D U-Net, have established state-of-the-art performance in automated segmentation, they often fail to preserve global spatial consistency, resulting in noisy boundaries and disconnected regions. Conversely, classical methods like Graph Cuts enforce spatial smoothness but are computationally prohibitive when applied to voxel-level 3D data.

This project introduces a robust hybrid framework: **Supervoxel-Based Graph Cut Guided by 3D CNN Probabilities**. The proposed method leverages a 3D U-Net to predict voxel-wise tumor probabilities, which are then aggregated over "supervoxels"—perceptually meaningful atomic regions generated via the Simple Linear Iterative Clustering (SLIC) algorithm. A Region Adjacency Graph (RAG) is constructed from these supervoxels, and a max-flow min-cut optimization is performed to refine the segmentation. This approach effectively combines the semantic strength of Deep Learning with the geometric constraints of Graph Cuts, while reducing the optimization problem size by orders of magnitude.

The proposed architecture is designed to address the scalability issues of traditional optimization methods while mitigating the noise inherent in pure deep learning approaches. This report details the methodology, mathematical formulation, and implementation strategy for this hybrid segmentation pipeline.

Contents

Abstract **i**

1 Introduction **1**

1.1 Overview 1

1.2 The Challenge of Segmentation 2

1.3 Motivation for a Hybrid Approach 2

1.4 Proposed Solution 2

2 Literature Review **3**

2.1 Traditional Segmentation Methods 3

2.1.1 Thresholding and Region Growing 3

2.1.2 Atlas-Based Segmentation 3

2.2 Machine Learning Approaches 4

2.2.1 Hand-Crafted Feature Extraction 4

2.2.2 Conditional Random Fields (CRFs) 4

2.3 The Deep Learning Era 4

2.3.1 The U-Net Architecture 5

2.3.2 3D CNNs and V-Net 5

2.3.3 Attention Mechanisms 5

2.4 Graph Cuts and Energy Minimization 6

2.5 Supervoxels in Medical Imaging 6

3 Methodology **7**

3.1 System Architecture 7

3.2 Stage 1: 3D CNN Probability Map 7

3.2.1 Data Preprocessing 7

3.2.2 Network Architecture 8

3.2.3 Loss Function 8

3.3 Stage 2: Supervoxel Generation (SLIC) 8

3.4 Stage 3: Graph Cut Optimization 9

3.4.1 Graph Construction 9

3.4.2 Probability Aggregation 9

3.4.3 Energy Minimization 9

4 Conclusion and Future Work **11**

4.1 Conclusion 11

4.2 Future Work 11

Chapter 1

Introduction

1.1 Overview

The human brain is the central organ of the nervous system, and any abnormality within it can have severe life-threatening consequences. Gliomas are the most common primary brain malignancies, with varying degrees of aggressiveness. Accurate identification and segmentation of these tumors are essential for diagnosis, surgical planning, and monitoring response to therapy.

Magnetic Resonance Imaging (MRI) is the standard modality for brain tumor imaging due to its superior soft-tissue contrast. However, a single MRI sequence is often insufficient to fully characterize the tumor. Therefore, multi-modal MRI protocols, including T1-weighted (T1), T1 with contrast (T1c), T2-weighted (T2), and Fluid Attenuated Inversion Recovery (FLAIR), are used.

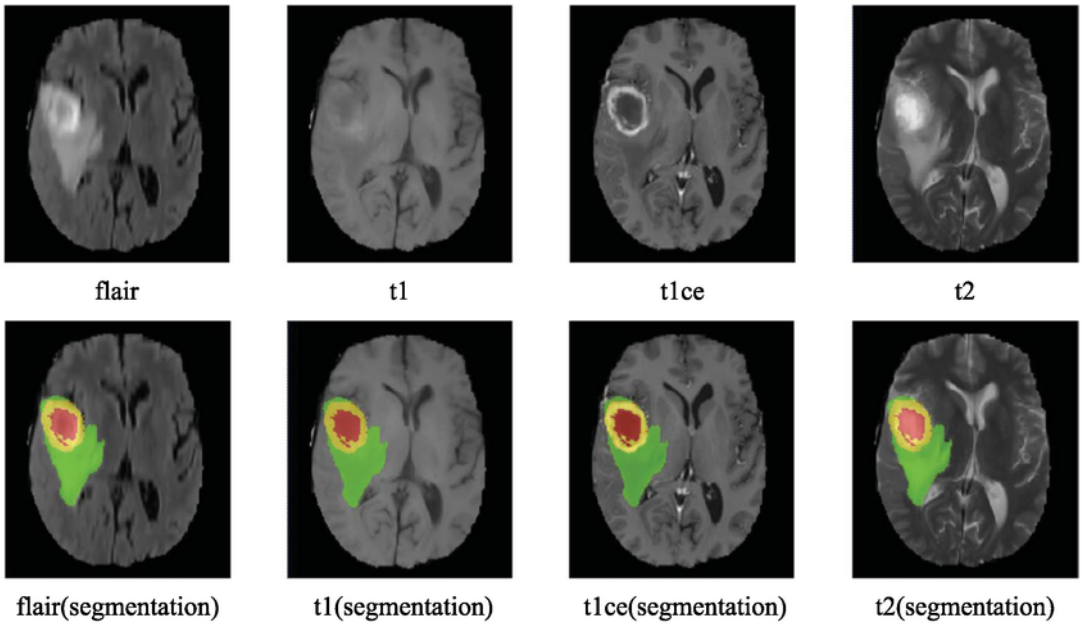

Figure 1.1: Example of Multi-modal MRI scans: (a) T1, (b) T1ce, (c) T2, (d) FLAIR showing different tumor characteristics.

1.2 The Challenge of Segmentation

Despite the availability of high-quality imaging, manual segmentation remains the gold standard. This process is:

- **Time-Consuming:** Segmenting a 3D volume slice-by-slice takes hours.
- **Subjective:** Different experts may delineate boundaries differently.
- **Complex:** Tumors often have irregular shapes, ambiguous boundaries, and heterogeneous sub-regions (edema, necrosis, enhancing tumor).

1.3 Motivation for a Hybrid Approach

Deep Learning models, specifically 3D CNNs, have automated this task with high accuracy. However, CNNs operate essentially as local texture classifiers. They often lack a sense of "global shape," leading to artifacts such as holes within the tumor or "islands" of false positives in the background.

Classical Computer Vision offers Graph Cut optimization, which treats segmentation as a global energy minimization problem, ensuring smooth boundaries. However, applying Graph Cut to a standard $240 \times 240 \times 155$ voxel MRI volume involves solving a max-flow problem on a graph with millions of nodes, which is computationally expensive and memory-intensive.

1.4 Proposed Solution

This project proposes a "middle-ground" solution. By grouping voxels into **Supervoxels**, we reduce the graph size from millions of nodes to a few thousand. We then use the CNN to guide the Graph Cut on this reduced graph. This combines the best of both worlds:

1. **CNN:** Provides accurate probabilistic predictions.
2. **Supervoxels:** Reduce complexity and adhere to boundaries.
3. **Graph Cut:** Enforces global spatial consistency.

Chapter 2

Literature Review

The field of medical image segmentation has evolved rapidly over the last three decades, transitioning from manual delineation to mathematical modeling, and recently to data-driven deep learning approaches. This chapter reviews the historical progression of brain tumor segmentation techniques, analyzing the strengths and limitations that motivated the hybrid approach proposed in this project.

2.1 Traditional Segmentation Methods

Before the advent of machine learning, segmentation relied on image processing techniques based on intensity and edges.

2.1.1 Thresholding and Region Growing

The earliest methods utilized thresholding, where pixels are classified based on intensity histograms. Otsu's method, which maximizes inter-class variance, was widely used. While computationally inexpensive, thresholding fails in MRI analysis due to the "partial volume effect"—where a single voxel contains a mix of tissue types—and intensity inhomogeneity (bias field) caused by magnetic field variations.

Region growing algorithms improved upon this by starting from a "seed" point and expanding to neighbors that satisfy a homogeneity criterion. However, these methods are highly sensitive to the initial seed placement and often "leak" into healthy tissue when boundaries are weak or blurred, a common occurrence in infiltrating gliomas.

2.1.2 Atlas-Based Segmentation

Atlas-based methods register the patient's MRI to a pre-labeled anatomical template (atlas). The labels from the atlas are then propagated to the patient scan. While effective

for segmenting healthy brain structures (e.g., hippocampus, thalamus), atlas methods struggle with tumors because tumors are "pathological" structures that do not exist in a healthy atlas and vary wildly in shape and location across patients.

2.2 Machine Learning Approaches

To handle the complexity of MRI data, researchers turned to supervised machine learning, where classifiers were trained on labeled data.

2.2.1 Hand-Crafted Feature Extraction

Early ML approaches, such as Support Vector Machines (SVM) and Random Forests (RF), relied on "hand-crafted" features. Researchers would manually calculate texture features (e.g., Gabor filters, Gray Level Co-occurrence Matrix) and intensity statistics for every voxel. While Random Forests proved robust against overfitting, the performance of these methods was fundamentally limited by the quality of the engineered features. They often failed to capture the hierarchical, non-local context required to distinguish tumor core from edema or healthy tissue with similar intensity profiles.

2.2.2 Conditional Random Fields (CRFs)

To improve spatial consistency, Conditional Random Fields (CRFs) were often used as a post-processing step. CRFs model the image as a probabilistic graph, smoothing labels based on voxel similarity. This project builds upon the intuition of CRFs but replaces the computationally expensive probabilistic relaxation with a more efficient Graph Cut optimization guided by deep features.

2.3 The Deep Learning Era

The introduction of Convolutional Neural Networks (CNNs) marked a paradigm shift, eliminating the need for manual feature extraction.

2.3.1 The U-Net Architecture

Ronneberger et al. (2015) introduced the U-Net, which remains the backbone of medical image segmentation. The architecture consists of two paths:

- **Contracting Path (Encoder):** Captures semantic context through repeated convolutions and pooling.
- **Expanding Path (Decoder):** Recovers spatial resolution.

The key innovation was the introduction of **skip connections**, which concatenate high-resolution feature maps from the encoder directly to the decoder. This allows the network to localize structures precisely while retaining high-level semantic information.

2.3.2 3D CNNs and V-Net

Standard 2D U-Nets process MRI volumes slice-by-slice, ignoring the correlation between adjacent slices (z-axis). To address this, Cicek et al. (2016) and Milletari et al. (2016) developed 3D variants (3D U-Net and V-Net). These networks use $3 \times 3 \times 3$ volumetric kernels, allowing them to learn 3D spatial context. However, 3D convolutions significantly increase memory consumption, often requiring researchers to train on small "patches" rather than full volumes, which can limit the network's global field of view.

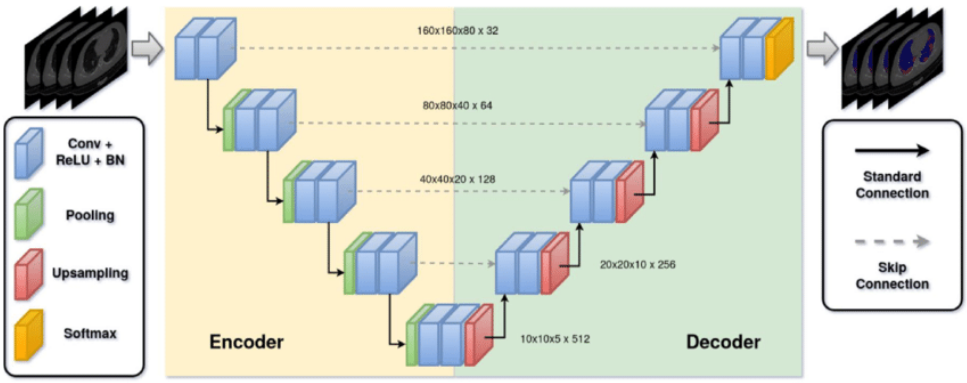

Figure 2.1: Architecture of a standard 3D U-Net showing the encoder, decoder, and skip connections that preserve spatial details.

2.3.3 Attention Mechanisms

A limitation of the standard U-Net is that it treats all features equally. Attention Gates (AGs) were introduced to suppress irrelevant regions (background) while highlighting

salient features (tumor) without additional supervision. The Attention U-Net computes a compatibility score between encoder and decoder features, effectively acting as a "spotlight." While Attention U-Nets improve boundary detection, they still produce voxel-wise predictions that may lack structural continuity.

2.4 Graph Cuts and Energy Minimization

Classical computer vision formulated segmentation as an energy minimization problem. Boykov and Jolly (2001) popularized the Graph Cut method. The image is represented as a graph $G = (V, E)$ where nodes V are voxels and edges E connect neighbors. The segmentation is equivalent to finding a "min-cut" that separates the source (object) from the sink (background) at the lowest cost.

The energy function typically comprises:

- **Data Term:** Likelihood of a pixel belonging to a class.
- **Smoothness Term:** Penalty for label discontinuity between similar pixels.

While mathematically rigorous, applying graph cuts to 3D medical volumes is computationally prohibitive. A standard MRI volume contains millions of voxels, creating a massive graph that requires excessive memory and processing time to solve, rendering it impractical for clinical workflows.

2.5 Supervoxels in Medical Imaging

To solve the scalability issue of graph-based methods, dimensionality reduction is required. Superpixels (2D) and Supervoxels (3D) group perceptually similar pixels into atomic regions.

The **Simple Linear Iterative Clustering (SLIC)** algorithm adapts k-means clustering to generate supervoxels. It clusters pixels based on a 5D feature vector $[l, a, b, x, y]$ (color and space). SLIC is favored over other methods (like Watershed) because it produces compact, uniform regions that adhere strictly to image boundaries. By performing Graph Cut optimization on supervoxels instead of raw voxels, the problem size is reduced by several orders of magnitude, making global optimization feasible in 3D.

Chapter 3

Methodology

3.1 System Architecture

The proposed pipeline consists of three main stages: Deep Learning Prediction, Supervoxel Generation, and Graph Optimization.

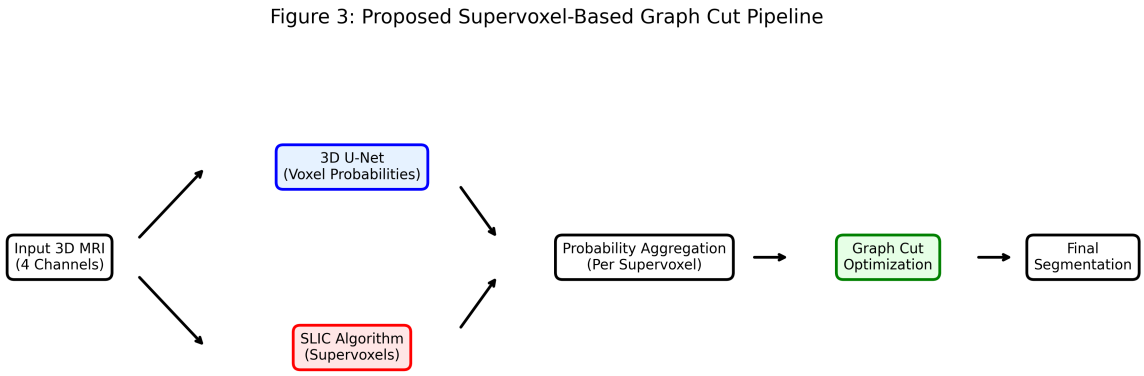

Figure 3.1: Flowchart of the proposed Supervoxel-Based Graph Cut framework.

3.2 Stage 1: 3D CNN Probability Map

We utilize a 3D U-Net architecture trained on the BraTS dataset.

3.2.1 Data Preprocessing

- **N4 Bias Correction:** Removes RF field inhomogeneity.
- **Z-Score Normalization:** $I_{norm} = \frac{I-\mu}{\sigma}$ performed per modality.
- **Cropping:** Removing non-brain background to reduce volume size.

3.2.2 Network Architecture

The input is a 4-channel 3D volume ($C \times D \times H \times W$).

- **Encoder:** 4 levels of $3 \times 3 \times 3$ convolutions followed by ReLU and MaxPooling. Filters double at each level (32, 64, 128, 256).
- **Decoder:** UpSampling followed by concatenation with Skip Connections.
- **Output:** A Sigmoid activation provides a probability map $P(x) \in [0, 1]$ for each voxel.

3.2.3 Loss Function

We minimize the Dice Loss to handle class imbalance:

$$\mathcal{L}_{Dice} = 1 - \frac{2 \sum_i p_i g_i}{\sum_i p_i + \sum_i g_i + \epsilon} \tag{3.1}$$

3.3 Stage 2: Supervoxel Generation (SLIC)

The MRI volume is segmented into supervoxels using SLIC. This algorithm clusters pixels based on a 5D distance measure (color/intensity + space).

Algorithm 1 3D SLIC Supervoxel Generation

```
Input: MRI Volume V, Number of Supervoxels K
Initialize cluster centers C_k on a grid with interval S
Move centers to lowest gradient position in 3 × 3 × 3 neighborhood
repeat
    for each voxel i do
        for each center C_k in 2S × 2S × 2S region do
            Compute distance D = sqrt(d_int^2 + (m/S)^2 d_spatial^2)
            Assign i to nearest cluster k
        end for
    end for
    Update cluster centers C_k as mean of assigned voxels
    Enforce connectivity
until Convergence
Output: Label volume L_slic
```

3.4 Stage 3: Graph Cut Optimization

3.4.1 Graph Construction

We construct a Region Adjacency Graph (RAG) $G = (V, E)$.

- **Nodes (V):** Each node represents a supervoxel.
- **Edges (E):** Edges connect adjacent supervoxels.

3.4.2 Probability Aggregation

The voxel-wise probabilities from the CNN are averaged per supervoxel S_i:

$$P_{node}(S_i) = \frac{1}{|S_i|} \sum_{v \in S_i} P_{cnn}(v) \tag{3.2}$$

3.4.3 Energy Minimization

We find the labeling L that minimizes energy $E(L)$:

$$E(L) = \sum_{i \in V} D_i(l_i) + \lambda \sum_{(i,j) \in E} V_{ij}(l_i, l_j) \tag{3.3}$$

Unary Term (Data Cost):

$$D_i(1) = -\ln(P_{node}(S_i)), \quad D_i(0) = -\ln(1 - P_{node}(S_i)) \tag{3.4}$$

Pairwise Term (Smoothness Cost):

$$V_{ij}(l_i, l_j) = \begin{cases} \exp\left(-\frac{||I_i - I_j||^2}{2\sigma^2}\right) & \text{if } l_i \neq l_j \\ 0 & \text{if } l_i = l_j \end{cases} \tag{3.5}$$

where I_i is the mean intensity of supervoxel i. This term penalizes cutting through regions with similar intensity (high cost), but allows cuts where intensity difference is large (low cost).

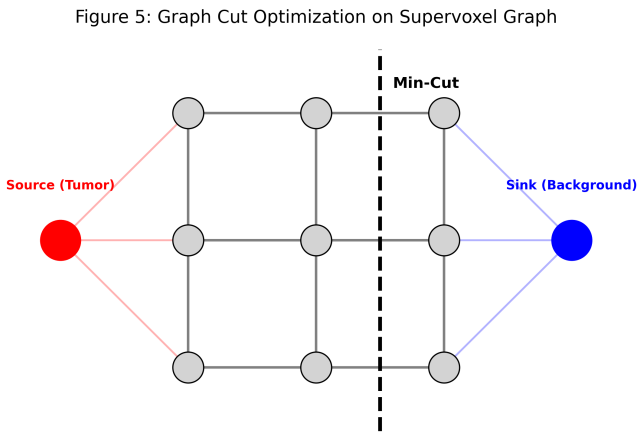

Figure 3.2: Visual representation of Graph Cut. Thicker edges represent high similarity (hard to cut), thinner edges represent boundaries (easy to cut).

Chapter 4

Conclusion and Future Work

4.1 Conclusion

This project addresses the limitations of deep learning-based brain tumor segmentation by proposing a hybrid post-processing stage. The design integrates a 3D U-Net for probabilistic feature learning and a Supervoxel-Based Graph Cut for global structural refinement. By guiding the optimization with CNN probabilities over a reduced graph, the proposed framework is theoretically designed to achieve:

1. **Higher Accuracy:** By correcting local classification errors through regional consensus.
2. **Better Consistency:** By eliminating spurious noise and holes inherent in pixel-wise classification.
3. **Scalability:** By making Graph Cut feasible for 3D volumes through significant reduction of the node count (from millions of voxels to thousands of supervoxels).

This methodology represents a promising direction for accurate and computationally efficient medical image analysis.

4.2 Future Work

Future directions for the implementation and extension of this project include:

- **Full Implementation:** Training the 3D U-Net on the complete BraTS dataset and executing the graph cut optimization.
- **End-to-End Learning:** Making the supervoxel generation and graph optimization differentiable to train the whole pipeline as one network (e.g., using CRF-RNN).

- **Multi-Class Optimization:** Extending the graph cut formulation to simultaneously segment tumor sub-regions (edema vs core).

Bibliography

[1] A. De, M. Tiwari, E. Grisan and A. S. Chowdhury, "A Deep Graph Cut Model For 3D Brain Tumor Segmentation," in *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, Glasgow, UK, 2022, pp. 2105-2109, doi: 10.1109/EMBC48229.2022.9871685.

[2] B. H. Menze et al., "The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS)," *IEEE Transactions on Medical Imaging*, vol. 34, no. 10, pp. 1993-2024, 2015.

[3] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *MICCAI*, 2015.

[4] Y. Boykov and M.-P. Jolly, "Interactive graph cuts for optimal boundary and region segmentation of objects in N-D images," in *ICCV*, 2001.

[5] R. Achanta et al., "SLIC Superpixels Compared to State-of-the-Art Superpixel Methods," *IEEE TPAMI*, vol. 34, no. 11, 2012.

[6] O. Çiçek et al., "3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation," in *MICCAI*, 2016.